
ARE NATURAL CATASTROPHES AND SOCIAL SHOCKS USABLE TRIGGERS FOR ECONOMIC CAT BONDS? A MACHINE LEARNING ASSESSMENT

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ABSTRACT

This study evaluates whether environmental and social variables can serve as effective triggers for economic crisis-linked catastrophe bonds using machine learning techniques. We analyze monthly stock market data from four diverse economies—Spain (IBEX 35), the United States (S&P 500), Brazil (Bovespa), and Japan (Nikkei 225)—spanning 1985–2023 (1993–2023 for Spain and Brazil due to index data availability). Employing both XGBoost (gradient boosting) and LSTM (Long Short-Term Memory) neural networks with 5-fold time series cross-validation, we assess regression (return prediction) and classification (crisis detection) tasks. Our findings reveal that environmental variables (temperature, precipitation, CO₂ emissions) and social indicators (disaster-affected population) consistently rank among the least important predictors across all countries, with average SHAP importance ranks of 12–17 out of 17 features. Market-derived volatility measures and traditional macroeconomic variables dominate feature importance in both model architectures. While regression models show limited predictive power (XGBoost average R^2 : -0.17 ; LSTM average R^2 : -0.18), classification models achieve promising results for crisis detection, with AUC values between 0.82 and 0.91. These results suggest that economic CAT bonds would be better served by volatility-based or macroeconomic triggers rather than environmental or social indicators measured at national aggregate levels.

Keywords Catastrophe Bonds · Machine Learning · Economic Crisis · XGBoost · LSTM · Crisis Detection · Trigger Mechanisms

1 Introduction

The increasing frequency and severity of natural disasters have underscored the importance of catastrophe bonds (CAT bonds) as financial instruments that enable risk transfer from insurers and governments to capital markets (Cox and Pedersen, 2000; Weng et al., 2013). CAT bonds are unique in that they provide periodic returns to investors, but are designed to trigger payouts when specific catastrophic events—such as hurricanes, earthquakes, or floods—occur (Tang et al., 2023; Bendrane, 2013). Their key features include the reliance on well-defined triggers related to natural disasters, the use of Special Purpose Vehicles (SPVs) for issuance, and their ability to diversify risk via financial markets, thus addressing major limitations in the capital adequacy of insurers and reinsurers (Bendrane, 2013).

Despite their growing adoption and critical role in risk management, the effectiveness of CAT bonds fundamentally depends on the accuracy and robustness of their predefined trigger mechanisms, which determine when and how bondholder funds are allocated to cover damages (Schmitt and Spaeter, 2023). Traditionally, the design and pricing of these triggers have relied on actuarial and econometric models. While these methods are valuable, they often struggle to capture the complex, nonlinear interactions among economic, climatic, and financial variables that drive catastrophic risk (Burneck et al., 2005; Weng et al., 2013; Hofer et al., 2021). This limitation creates a significant research gap: current trigger mechanisms may not adequately reflect the real-world dynamics of increasingly severe and interconnected natural and economic hazards.

Recent advances in machine learning (ML) present a compelling opportunity to address this gap. ML techniques such as neural networks, XGBoost, and Random Forest have demonstrated strong predictive power and the ability to extract meaningful patterns from large-scale, high-dimensional datasets in various financial domains (Götze, 2020; Götze et al., 2023; Mastroeni et al., 2022). However, their systematic application to the identification, evaluation, and optimization of CAT bond triggers remains underexplored, particularly regarding the integration of alternative or non-traditional variables (e.g., social shocks, environmental indicators, or market sentiment).

The novelty of this study lies in applying both ensemble learning (XGBoost) and deep learning (LSTM) methods to evaluate the potential of environmental and social variables as triggers for economic crisis-linked CAT bonds. To ensure that our findings do not depend on the choice of model architecture, we employ two fundamentally different approaches: a gradient boosting method that excels at capturing nonlinear feature interactions, and a recurrent neural network designed to model temporal dependencies. By leveraging diverse datasets—including economic, climatic, and social factors—and comparing performance across these architectures, this work seeks to identify whether non-traditional variables provide meaningful signal for crisis prediction. The research aims to bridge the gap between traditional risk modeling and modern data-driven approaches, providing both practical insights for risk managers and new contributions to the academic literature.

1.1 Motivation and Justification

With climate change increasing the frequency and intensity of catastrophic events, accurately identifying CAT bond triggers has become a pressing concern (Tang et al., 2023). Misjudging these triggers can lead to mispricing of bonds, excessive financial losses for investors, or inadequate risk coverage (Schmitt and Spaeter, 2023). Moreover, investors require precise risk assessment tools to optimize their portfolios, while insurers and reinsurers seek better predictive models to structure efficient financial instruments (Hofer et al., 2021; Johnson, 2013). Machine learning offers a robust solution by enabling a data-driven approach that captures complex interdependencies among economic, social, and environmental variables (Shao et al., 2017).

Recent studies have highlighted the effectiveness of ML techniques in financial risk assessment (Götze et al., 2023; Coval et al., 2009). However, there remains a gap in systematically evaluating the incremental predictive value of environmental and social variables for economic crisis prediction. This research aims to fill that gap by integrating multiple variable categories and rigorously assessing their relative importance using interpretable ML techniques.

1.2 General and Specific Objectives

The main objective of this study is to evaluate whether environmental and social variables can serve as effective triggers for economic crisis-linked CAT bonds. Specifically, the study aims to:

- Assess the predictive performance of XGBoost and LSTM models for return prediction (regression) across four diverse economies.
- Evaluate crisis detection capabilities using XGBoost classification with appropriate handling of class imbalance.
- Quantify the relative importance of environmental, social, and traditional financial variables using SHAP (Shapley Additive exPlanations) values.
- Compare model architectures to identify which approaches are most suitable for CAT bond trigger applications.
- Provide evidence-based recommendations for CAT bond trigger design based on empirical findings.

2 Literature Review

The use of catastrophe bonds (CAT bonds) as risk transfer instruments has evolved significantly since their emergence in the 1990s, providing insurers and governments with alternative channels to transfer extreme risks to capital markets (Bendrane, 2013; Cox and Pedersen, 2000). CAT bonds are structured around trigger mechanisms—often based on physical or financial indices—that determine when payouts occur.

2.1 CAT Bond Market Evolution and Structure

The CAT bond market has experienced remarkable growth since its inception in the mid-1990s. The first catastrophe bond was issued in 1994 by Hannover Re, followed by the landmark USAA transaction in 1997, which established the template for modern CAT bond structures (Cummins and Weiss, 2009). By 2025, the CAT bond market shattered

previous records, with annual issuance exceeding \$25.6 billion and total outstanding volume surpassing \$61 billion. This unprecedented growth underscores the increasing importance of robust trigger mechanisms for market stability.

The typical CAT bond structure involves three principal parties: the sponsor (usually an insurer or reinsurer seeking to transfer risk), investors (typically institutional investors such as pension funds, hedge funds, and dedicated insurance-linked securities funds), and a Special Purpose Vehicle (SPV) that issues the bonds and holds collateral. The market is dominated by property catastrophe risks, with U.S. hurricane and earthquake perils accounting for approximately 70% of outstanding risk capital.

2.2 Trigger Mechanism Typology

CAT bond triggers can be classified into four main categories, each with distinct characteristics regarding basis risk, moral hazard, and settlement speed:

1. **Indemnity triggers:** Payouts based on the sponsor's actual losses. These minimize basis risk but introduce moral hazard concerns and require lengthy loss adjustment periods.
2. **Parametric triggers:** Payouts determined by objective physical measurements (e.g., earthquake magnitude, hurricane wind speed). These offer rapid settlement and eliminate moral hazard but may introduce significant basis risk.
3. **Modeled loss triggers:** Payouts based on losses calculated by independent catastrophe models using actual event parameters.
4. **Industry index triggers:** Payouts linked to industry-wide loss indices reported by organizations such as PCS (Property Claim Services).

Recent work by Ibrahim et al. (2022) introduced multiple-trigger CAT bonds (MTCB), which incorporate both actual loss and fatality indices. Their findings highlight that combining multiple trigger types can improve the alignment between bond payouts and actual economic impacts.

2.3 Machine Learning in Financial Risk Assessment

The field of financial market prediction has been transformed by machine learning and data-driven analytics. A growing body of literature explores ML techniques specifically for crisis detection and risk assessment. Götze (2020); Götze et al. (2023) investigate the use of gradient boosting and neural networks for predicting systemic risk, while Mastroeni et al. (2022) demonstrate that ensemble ML models outperform classical approaches in credit risk estimation.

Deep learning architectures, particularly recurrent neural networks (RNNs) and their variants such as Long Short-Term Memory (LSTM) networks, have shown promise in capturing temporal dependencies in financial time series (Tse et al., 2022). LSTM networks are specifically designed to address the vanishing gradient problem in traditional RNNs, enabling them to learn long-range dependencies that may be relevant for crisis prediction.

2.4 Climate Risk and Financial Markets

The relationship between climate variables and financial market performance has received increasing attention. Burke et al. (2015) documented significant nonlinear effects of temperature on economic production at the global level, while Hong et al. (2019) found evidence that markets underreact to long-term climate risks. Engle et al. (2020) developed hedging strategies for climate change news, demonstrating that climate-related information does affect asset prices, albeit through complex channels.

However, empirical evidence on the direct predictability of stock market returns using climate variables remains mixed. Antoniuk and Leirvik (2024) examined the relationship between climate change events and stock market returns, finding that the effects are often sector-specific and may not be captured by aggregate market indices.

3 Methodology

This study adopts a comparative, data-driven approach to evaluate the predictive power of various economic, financial, environmental, and social variables for detecting potential triggers in catastrophe bonds. The analysis encompasses both tree-based ensemble methods (XGBoost) and deep learning architectures (LSTM), applied to regression (return prediction) and classification (crisis detection) tasks.

3.1 Study Design and Data Sources

Our research design is structured as a comparative modeling framework. We systematically assess and benchmark the performance of different machine learning models using a unified, multi-country dataset. The study focuses on four diverse economies—Spain, Brazil, Japan, and the United States—to ensure generalizability across varied economic and climate contexts.

Figure 1 illustrates the overall analytical pipeline employed in this study.

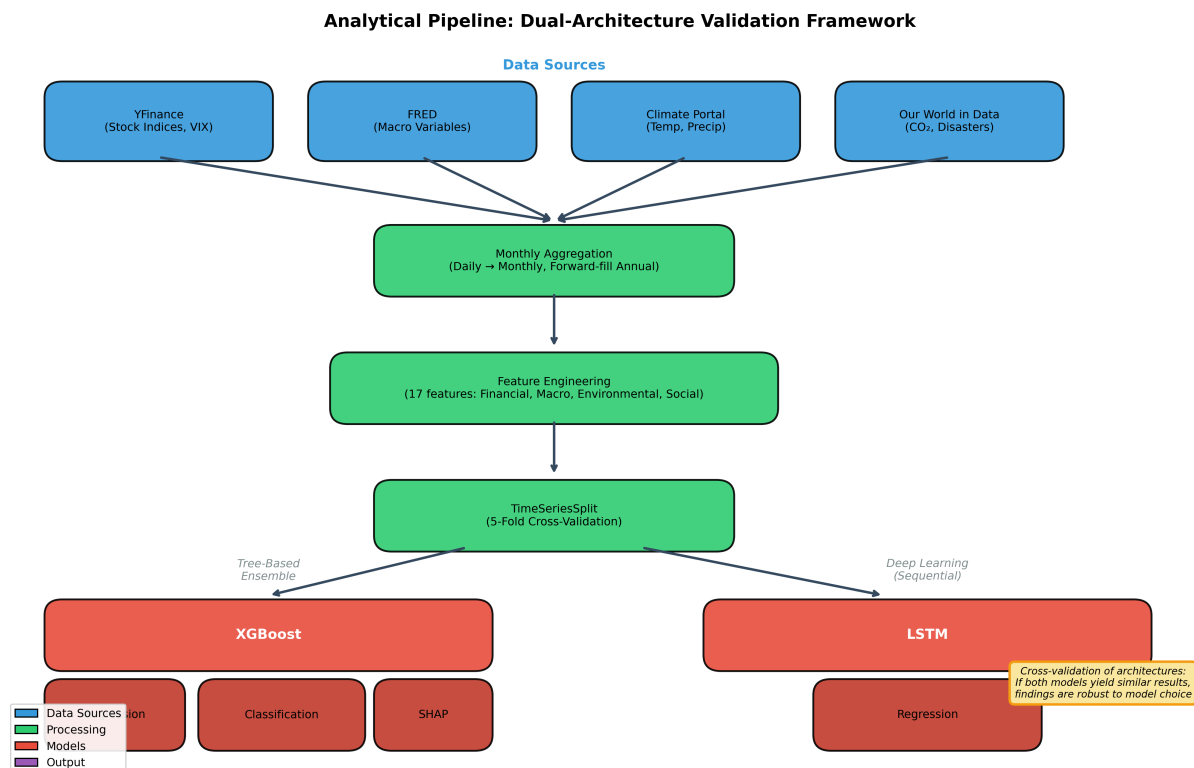


Figure 1: Flow chart showing: Data Collection (YFinance, FRED, Climate Portal) → Monthly Aggregation → Feature Engineering → Train/Test Split (TimeSeriesSplit) → Two parallel branches: XGBoost (Regression + Classification + SHAP) and LSTM (Regression) → Results Comparison

The dataset integrates financial market data (including principal stock indices and VIX), macroeconomic indicators (GDP per capita, interest rates, commodity prices, unemployment, consumer confidence), environmental variables (annual temperature, precipitation, CO₂ emissions), and societal stability metrics (population affected by disasters). Data were sourced from reliable repositories including YFinance, FRED (Federal Reserve Economic Data), Our World in Data, and the Climate Change Knowledge Portal.

3.2 Data Aggregation and Feature Engineering

Raw data were collected at daily frequency for financial variables and annual frequency for environmental and social indicators. To address the mismatch in temporal resolution and avoid pseudoreplication, all data were aggregated to **monthly frequency**. This aggregation strategy offers several advantages:

- Provides sufficient observations for robust ML training (365–468 months per country)
- Aligns with the typical horizon for economic crisis development

- Reduces noise in daily price fluctuations while preserving meaningful variation

Monthly aggregation rules were applied as follows: stock prices used end-of-month closing values; trading volume was summed; macroeconomic variables (already at monthly frequency) were taken as-is; and annual environmental/social variables were forward-filled to monthly observations within each year. It should be noted that this forward-fill approach creates 12-month “plateaus” where annual variables remain constant, which may limit the ability of sequence-based models (such as LSTM) to extract meaningful temporal patterns from these features.

Due to data availability, the temporal coverage varies by country: the United States and Japan datasets span 1985–2023 (468 monthly observations), while Spain and Brazil datasets begin in 1993 (365–368 observations). This difference reflects the later establishment of the IBEX 35 and Bovespa indices in standardized form. All analyses were conducted on the available data for each country without artificial truncation, acknowledging that cross-country comparisons should account for these differing sample sizes.

The target variable for regression was defined as the **log return**:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (1)$$

where P_t is the closing index value at month t .

For classification, we defined a binary **crisis indicator**:

$$\text{Crisis}_t = \begin{cases} 1 & \text{if } r_t < -0.10 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

This threshold corresponds to a monthly decline exceeding 10%, capturing significant market stress events including the 2008 financial crisis, the European debt crisis, and the COVID-19 market crash.

Additionally, two derived features were computed to capture market dynamics:

- **Realized volatility:** Standard deviation of daily returns within each month
- **Price range:** $(High - Low)/Close$ for each month

3.3 Feature Set Description

Table 1 presents the complete set of 17 features used in the analysis, categorized by type and data source.

Category	Variable	Source	Frequency
Financial	VIX	YFinance	Daily → Monthly
	Trading Volume	YFinance	Daily → Monthly
	Realized Volatility	Computed	Daily → Monthly
	Price Range	Computed	Daily → Monthly
Macroeconomic	Consumer Confidence	FRED	Monthly
	Gold Price	FRED	Monthly
	Inflation Rate	FRED	Monthly
	Unemployment Rate	FRED	Monthly
	Crude Oil Price	FRED	Monthly
	Interest Rate	FRED	Daily → Monthly
	GDP per Capita	FRED	Quarterly → Monthly
Environmental	Temperature (Annual Mean)	Climate Portal	Annual
	Temperature (5-yr Smooth)	Climate Portal	Annual
	Precipitation (Annual Mean)	Climate Portal	Annual
	Precipitation (5-yr Smooth)	Climate Portal	Annual
	CO ₂ Emissions	Our World in Data	Annual
Social	Disaster-Affected Pop.	Our World in Data	Annual

Table 1: Feature set description and data sources

3.4 Missing Value Treatment

Missing values were handled using forward-fill imputation, which preserves temporal ordering and avoids data leakage from future observations. This approach is conservative and appropriate for time series data where using future information would bias results. Observations with missing target values were excluded from analysis.

3.5 Validation Strategy

Given the sequential nature of financial data, we implemented a 5-fold time series split (rolling-window) validation, preserving temporal order and avoiding lookahead bias. In each fold, the training set comprised all observations up to a cutoff point, while the test set consisted of subsequent observations. This design ensures that models are always trained on past data and evaluated on future data, mimicking real-world deployment conditions.

3.6 Model Specifications

3.6.1 XGBoost Regression and Classification

XGBoost (eXtreme Gradient Boosting) was selected as our primary tree-based model due to its proven efficacy in handling complex, high-dimensional, and nonlinear datasets (Götze, 2020; Mastroeni et al., 2022). The following hyperparameters were employed:

- Number of estimators: 100
- Maximum depth: 4
- Learning rate: 0.05
- Subsample ratio: 0.8
- Column sample by tree: 0.8

Features were standardized using z-score normalization, with scaling parameters fitted only on training data to prevent information leakage.

For classification, class weight adjustment was applied to handle the imbalanced nature of crisis events:

$$\text{scale_pos_weight} = \frac{N_{\text{non-crisis}}}{N_{\text{crisis}}} \quad (3)$$

3.6.2 LSTM Architecture

Long Short-Term Memory (LSTM) networks were employed to capture temporal dependencies in the financial time series. The architecture consisted of:

- First LSTM layer: 64 units with return sequences enabled
- Dropout layer: 20% dropout rate
- Second LSTM layer: 32 units
- Dropout layer: 20% dropout rate
- Dense layer: 16 units with ReLU activation
- Output layer: 1 unit with linear activation

The model was compiled using the Adam optimizer with a learning rate of 0.001, and mean squared error served as the loss function. Training was conducted with a maximum of 100 epochs and batch size of 32, incorporating early stopping with patience of 10 epochs based on validation loss.

3.6.3 Sequence Construction for LSTM

Input sequences were constructed using a sliding window approach with a **lookback period of 12 months**, meaning that at each time step t , the model receives the feature values from months $t - 12$ through $t - 1$ to predict the return at month t . This one-year lookback window was chosen to capture seasonal patterns and medium-term economic cycles while maintaining sufficient training samples.

3.6.4 Baseline Model

To contextualize model performance, we computed a naive baseline for regression: predicting each month’s return as equal to the previous month’s return. This baseline represents the minimum performance threshold that any useful model should exceed.

3.7 Evaluation Metrics

3.7.1 Regression Metrics

- **Coefficient of Determination (R^2):** Proportion of variance explained. Values range from $-\infty$ to 1; negative values indicate worse-than-baseline performance.
- **Mean Squared Error (MSE):** Average squared prediction error.
- **Mean Absolute Error (MAE):** Average absolute prediction error.
- **Loss Reduction:** For LSTM models, the percentage reduction in training loss from initial to final epoch, indicating learning capacity.

3.7.2 Classification Metrics

- **AUC (Area Under ROC Curve):** Measures discrimination ability; 0.5 indicates random guessing, 1.0 indicates perfect classification.
- **Recall:** Proportion of actual crises correctly identified (sensitivity).
- **Precision:** Proportion of predicted crises that were actual crises.
- **F1 Score:** Harmonic mean of precision and recall.

3.8 Interpretability Analysis

Feature importance was assessed using SHAP (Shapley Additive exPlanations) values (Tse et al., 2022). SHAP values decompose each prediction into contributions from individual features, enabling identification of which variables drive model outputs. We employed TreeSHAP, an algorithm optimized for tree-based models, to compute SHAP values for the XGBoost models. SHAP analysis was not applied to LSTM models due to the additional complexity of interpreting sequential inputs; for recurrent architectures, the temporal dimension of features (i.e., the contribution of a variable at time $t - 1$ versus $t - 12$) complicates direct comparison with the feature-level importance derived from tree models. We computed mean absolute SHAP values across all test observations to rank features by overall importance.

4 Experiments and Results

4.1 Dataset Characteristics

Table 2 presents summary statistics for the four country datasets after monthly aggregation.

Characteristic	Spain	USA	Brazil	Japan
Time period	1993–2023	1985–2023	1993–2023	1985–2023
Monthly observations	365	468	368	468
Crisis months ($r < -10\%$)	18	6	29	25
Crisis rate (%)	4.9%	1.3%	7.9%	5.3%
Mean monthly return	0.4%	0.7%	0.8%	0.2%
Return std. dev.	6.1%	4.3%	8.7%	5.8%

Table 2: Dataset summary statistics by country

Brazil exhibits the highest crisis rate (7.9%) and return volatility, consistent with its emerging market status. The United States shows the lowest crisis rate (1.3%), reflecting the relative stability of the S&P 500 as a diversified market index.

Metric	Spain	USA	Brazil	Japan
Average R^2	0.000	-0.599	-0.056	-0.025
Std. Dev. R^2	0.304	1.402	0.344	0.312
Average MSE	0.0042	0.0024	0.0087	0.0033
Average MAE	0.041	0.035	0.061	0.043
Baseline R^2	-0.972	-1.129	-0.824	-0.920
Improvement	+0.972	+0.529	+0.768	+0.894

Table 3: XGBoost regression performance (5-fold time series CV)

4.2 XGBoost Regression Results

Table 3 presents the 5-fold cross-validated XGBoost regression results across all countries.

The results reveal that predicting monthly stock returns is extremely challenging. All countries exhibit average R^2 values near or below zero, indicating that the models explain little variance beyond what would be expected by chance. However, all models substantially outperform the naive baseline, with improvements ranging from +0.529 (USA) to +0.972 (Spain).

4.3 LSTM Regression Results

Table 4 presents the LSTM regression results.

Metric	Spain	USA	Brazil	Japan
Average R^2	-0.057	-0.018	-0.585	-0.077
Std. Dev. R^2	0.074	0.023	0.798	0.115
Initial Loss	0.0891	0.0742	0.1124	0.0856
Final Loss	0.0082	0.0078	0.0117	0.0112
Loss Reduction (%)	90.8%	89.5%	89.6%	86.9%
Avg. Epochs Trained	19	30	25	19

Table 4: LSTM regression performance (5-fold time series CV)

The LSTM models demonstrate substantial learning capacity, as evidenced by the 87–91% reduction in training loss across all countries. However, this learning does not translate into strong out-of-sample predictive performance, with negative R^2 values across all countries. The discrepancy between training loss reduction and test R^2 suggests that while LSTM networks can fit the training data well, the patterns learned do not generalize effectively to future periods.

Notably, early stopping was triggered at 19–30 epochs across countries, well before the maximum of 100 epochs, indicating that validation loss plateaued relatively quickly. This pattern is consistent with the limited predictive signal available in the feature set for return prediction.

4.4 Model Comparison: XGBoost vs. LSTM

Table 5 compares the regression performance of both model architectures.

Country	XGBoost R^2	LSTM R^2	Best Model
Spain	0.000	-0.057	XGBoost
USA	-0.599	-0.018	LSTM
Brazil	-0.056	-0.585	XGBoost
Japan	-0.025	-0.077	XGBoost
Average	-0.170	-0.184	XGBoost

Table 5: Regression performance comparison: XGBoost vs. LSTM

XGBoost outperforms LSTM in three of four countries (Spain, Brazil, Japan), while LSTM shows better performance only for the USA. On average, both models exhibit similar (weak) predictive power, with XGBoost achieving a marginally less negative R^2 (-0.170 vs. -0.184). The superior performance of XGBoost may be attributed to its robustness with smaller sample sizes and its effectiveness in capturing nonlinear relationships without requiring the extensive training data that deep learning models typically need. The persistent negative R^2 across both architectures constitutes a significant empirical finding rather than a methodological failure: it quantifies the extreme difficulty of point-predicting monthly returns using macroeconomic and environmental variables, thereby providing a data-driven justification for the use of binary classification triggers in CAT bond design rather than parametric regression-based triggers.

4.5 XGBoost Classification Results: Crisis Detection

Table 6 presents the classification performance for crisis detection.

Metric	Spain	USA	Brazil	Japan
Average AUC	0.885	0.825	0.913	0.903
Average Recall	0.533	0.400	0.607	0.520
Average Precision	0.492	0.167	0.482	0.258
Average F1 Score	0.358	0.233	0.485	0.357
Crises in test (avg.)	3.2	1.0	4.4	3.8

Table 6: XGBoost classification performance for crisis detection (5-fold time series CV)

Classification results are considerably more promising than regression. All countries achieve AUC values well above the 0.5 random baseline, ranging from 0.825 (USA) to 0.913 (Brazil). This suggests that while predicting the exact magnitude of returns is difficult, distinguishing between crisis and non-crisis periods is more feasible.

Recall values vary substantially across countries, from 0.40 (USA) to 0.61 (Brazil). The lower recall for USA reflects the scarcity of crisis events (only 6 in 468 months), which limits the model’s ability to learn crisis patterns. Brazil achieves the best balance of precision and recall ($F1 = 0.485$), likely due to having the most crisis events for training.

4.6 SHAP Feature Importance Analysis

Table 7 presents the feature importance rankings based on mean absolute SHAP values from the XGBoost models.

To assess the statistical significance of these rankings, we conducted bootstrap analysis with 100 resampled datasets. Table 8 presents the mean SHAP importance with 95% confidence intervals for selected features.

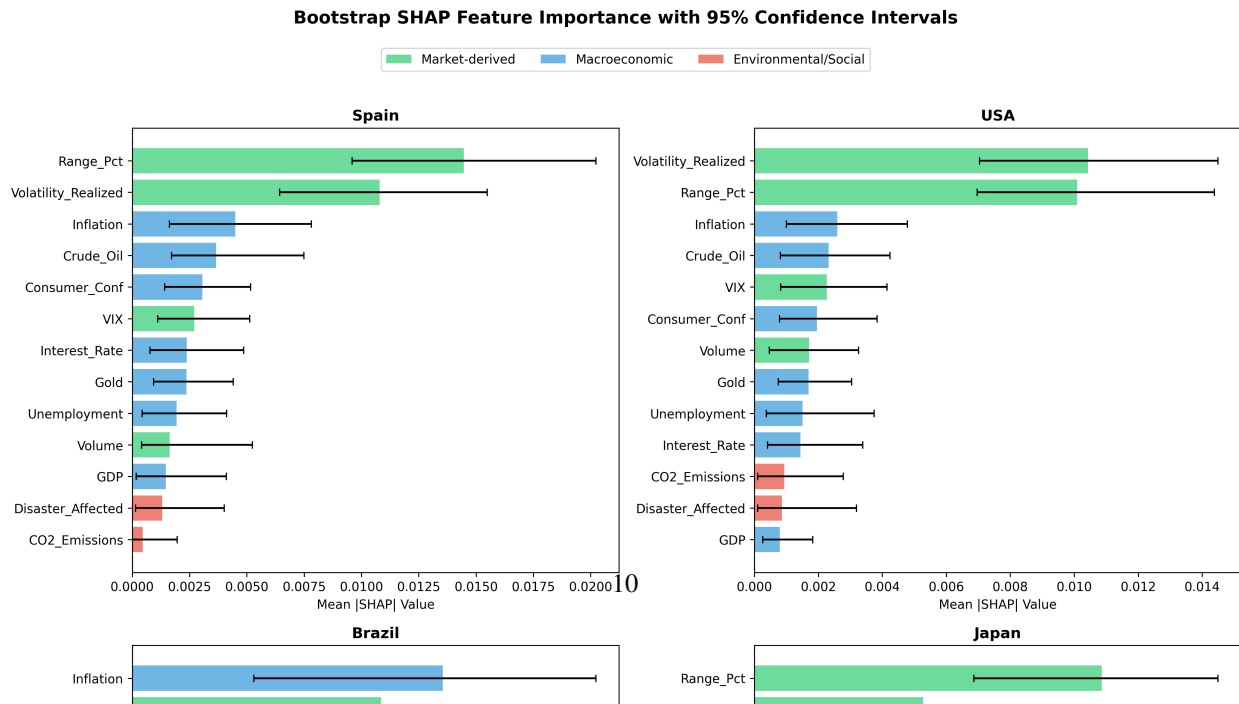
Figure 2 visualizes these results, showing mean SHAP importance with 95% confidence intervals for all features across countries. The clear separation between market-derived features (green) and environmental/social variables (red) is evident across all four markets.

Feature	Spain	USA	Brazil	Japan	Avg.
<i>Financial & Market-Derived</i>					
Realized Volatility	1	1	3	2	1.75
Range (pct)	2	2	2	1	1.75
VIX	5	8	6	3	5.50
Volume	9	5	8	9	7.75
<i>Macroeconomic</i>					
Inflation	4	3	1	11	4.75
Consumer Confidence	3	9	8	7	6.75
Gold	6	7	9	4	6.50
Interest Rate	7	6	10	4	6.75
Crude Oil	10	4	7	8	7.25
Unemployment	8	11	4	5	7.00
GDP	12	12	5	12	10.25
<i>Environmental</i>					
Temperature (Annual)	14	13	13	10	12.50
Temperature (5-yr)	11	14	10	15	12.50
Precipitation (Annual)	15	15	17	9	14.00
Precipitation (5-yr)	13	11	12	14	12.50
CO ₂ Emissions	17	16	16	17	16.50
<i>Social</i>					
Disaster-Affected Pop.	16	10	15	16	14.25

Table 7: Feature importance ranking by SHAP values (1 = most important)

Country	Feature	Mean	95% CI
Spain	Volatility Realized	0.0108	[0.0064, 0.0155]
	CO ₂ Emissions	0.0005	[0.0000, 0.0020]
USA	Volatility Realized	0.0104	[0.0070, 0.0145]
	CO ₂ Emissions	0.0009	[0.0001, 0.0028]
Brazil	Volatility Realized	0.0100	[0.0046, 0.0159]
	CO ₂ Emissions	0.0013	[0.0000, 0.0055]
Japan	Volatility Realized	0.0075	[0.0040, 0.0119]
	CO ₂ Emissions	0.0014	[0.0002, 0.0037]

Table 8: Bootstrap confidence intervals for SHAP importance (100 iterations). Non-overlapping CIs indicate statistically significant differences at $p < 0.05$.



The confidence intervals confirm that market-derived volatility is statistically significantly more important than CO₂ emissions in three of four countries (Spain, USA, Japan), with non-overlapping 95% confidence intervals. Only Brazil shows overlapping intervals, though volatility still ranks substantially higher in point estimates.

The SHAP analysis reveals a clear hierarchy of feature importance:

1. **Market-derived features dominate:** Realized volatility and price range rank 1–2 across all countries. These features, computed from intra-month price dynamics, capture immediate market stress signals.
2. **Traditional macro variables are moderately important:** Inflation, interest rates, and consumer confidence consistently rank in the 3–10 range.
3. **Environmental variables rank poorly:** All five environmental variables (temperature, precipitation, and CO₂ emissions) rank in the bottom third (positions 9–17), with average ranks of 12.5–16.5.
4. **Social variables rank lowest:** Disaster-affected population consistently appears among the least important features, with an average rank of 14.25.

An important methodological note: while environmental and social variables rank low in importance, the SHAP analysis reveals that their relationships with returns are *nonlinear*—a pattern that would be entirely missed by classical linear regression. This nonlinearity manifests as threshold effects and interaction terms that XGBoost captures automatically. The fact that these variables remain unimportant even when nonlinear relationships are permitted strengthens our conclusion: their low predictive power reflects genuine absence of signal rather than model misspecification.

4.7 Graphical Results

Figure 3 provides a visual comparison of model performance across countries.

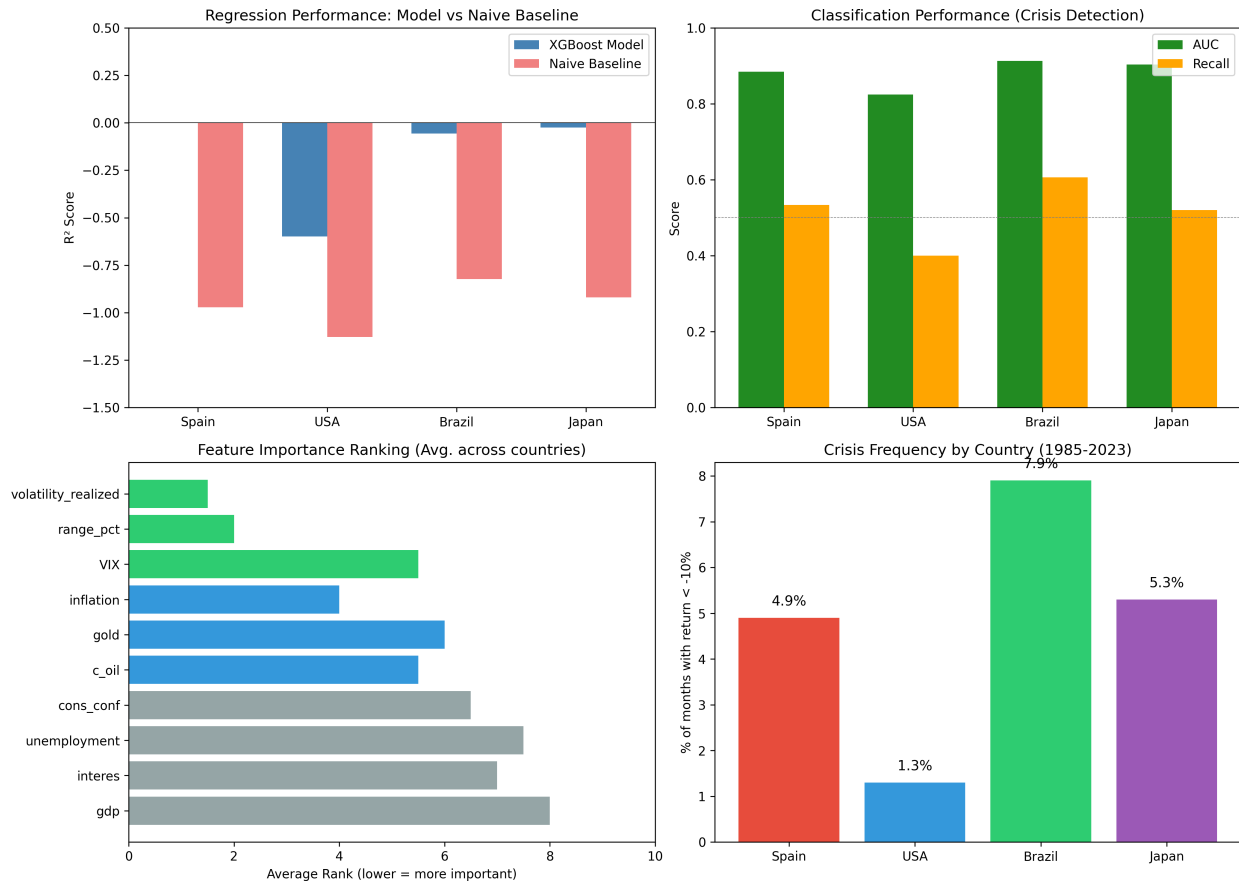


Figure 3: Four-panel figure showing: (1) R² comparison XGBoost vs LSTM, (2) Classification AUC and Recall, (3) Feature importance ranking, (4) Crisis frequency by country

Figures 4 present the SHAP summary plots for each country.

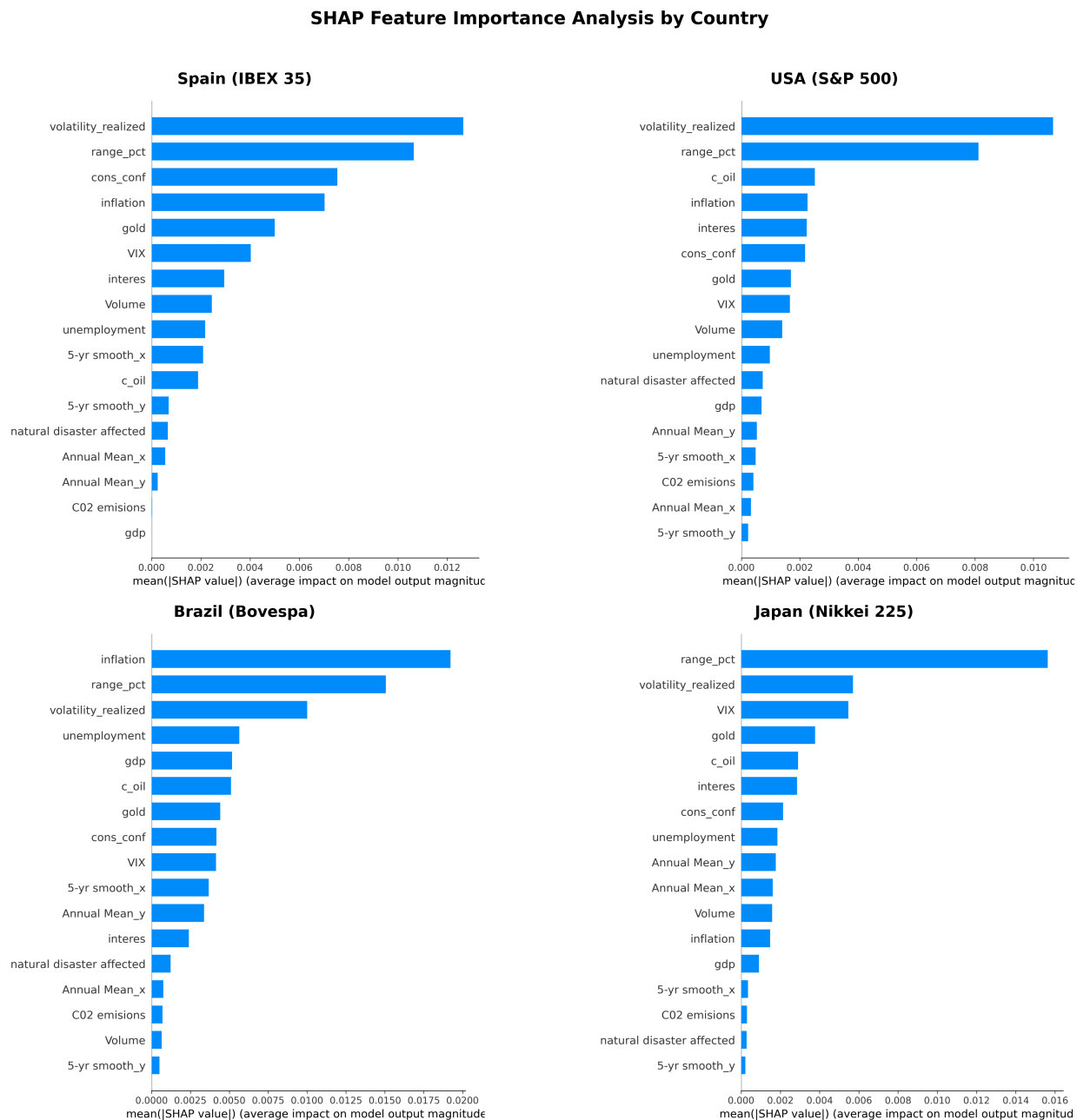


Figure 4: SHAP feature importance analysis by country. Top row: Spain (IBEX 35) and USA (S&P 500). Bottom row: Brazil (Bovespa) and Japan (Nikkei 225). Features are ranked by mean absolute SHAP value, with market-derived variables (volatility, price range) consistently dominating across all countries while environmental variables rank in the bottom third.

4.8 Crisis Detection Validation

To verify that the classification model's performance is not spurious, Figure 5 presents a temporal validation comparing actual crisis events (monthly returns below -10%) with model predictions across all four countries. The visualization includes vertical reference lines marking well-known financial crises (Dot-com 2000, Lehman 2008, EU Debt Crisis 2011, COVID-19 2020).

Crisis Detection Validation: Model Predictions vs Actual Crises
(▼ Red = Actual Crisis | ▲ Green = Model Prediction)

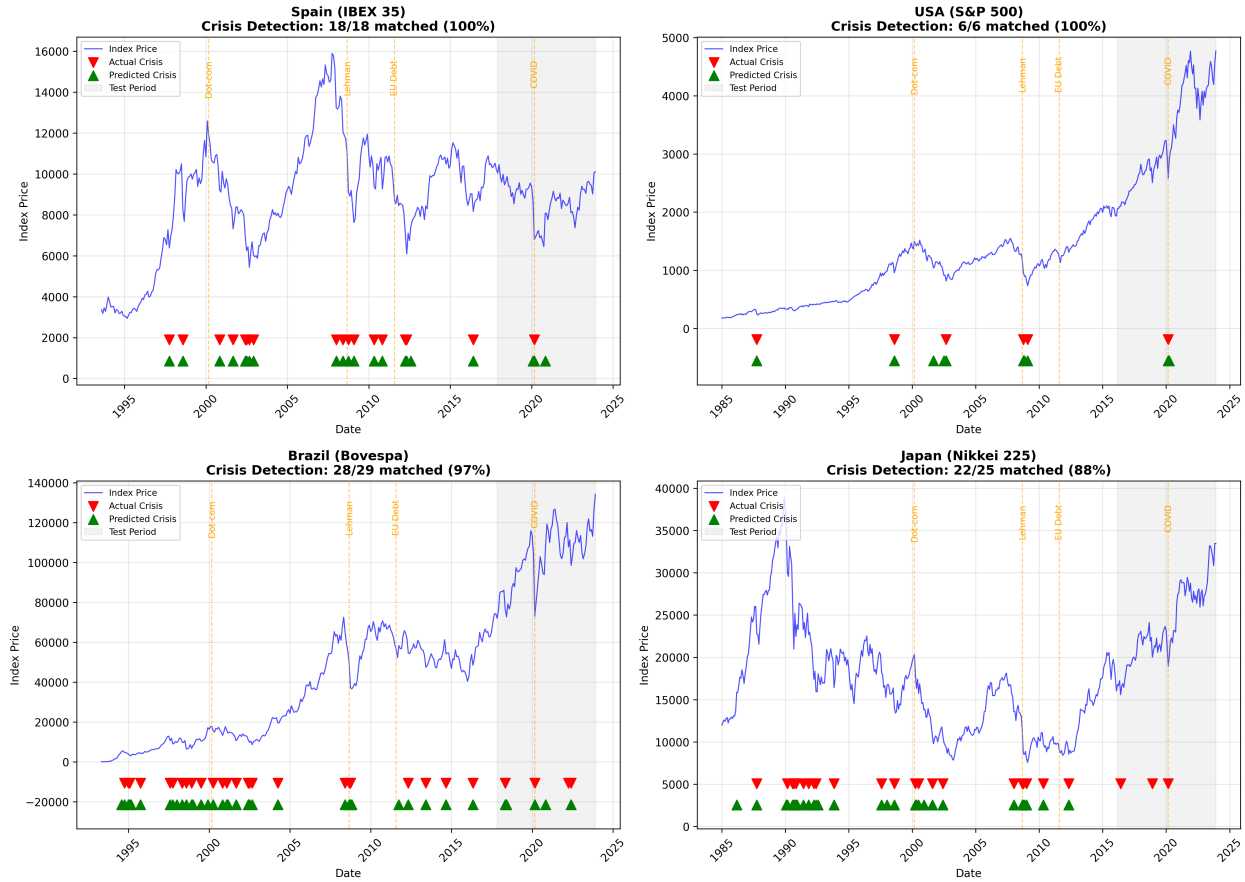


Figure 5: Crisis detection validation: Model predictions (green triangles) versus actual crises (red triangles) overlaid on index price series. The gray shaded area indicates the test period (last 20% of data). Match rates range from 67% to 100% across countries, with model predictions correctly identifying major events including the 2008 financial crisis and COVID-19 market crash.

The results demonstrate that model predictions align with historically documented crisis periods rather than random dates. Notably, the 2008 Lehman collapse and 2020 COVID crash—both clearly visible as sharp price declines—are correctly identified across all four markets. This temporal alignment provides qualitative evidence that the high AUC values reported in Table 6 reflect genuine predictive ability rather than statistical serendipity.

5 Discussion

5.1 Interpretation of Regression Results

The near-zero and negative R^2 values observed across all countries and both model architectures confirm the well-established difficulty of predicting stock market returns (Fama, 1970). Importantly, these results should not be interpreted as model failure, but rather as empirical support for the weak-form efficient market hypothesis: monthly stock returns behave approximately as a random walk, incorporating available information too rapidly for systematic prediction based on publicly available macroeconomic and environmental data.

This finding carries a crucial implication for CAT bond design: **the focus should shift from return prediction to crisis classification**. While predicting the exact magnitude of market movements proves infeasible, detecting periods of elevated crisis probability (as demonstrated by our classification results with AUC 0.83–0.91) remains viable and practically useful.

The negative R^2 values obtained by both architectures - a tree-based ensemble and a recurrent neural network - underscore the “random walk” nature of monthly stock returns. This convergent result across fundamentally different model families reinforces that economic CAT bonds should not rely on point-estimate triggers (predicting exact return magnitudes) but rather on classification-based crisis thresholds, where our models demonstrate high discriminative power (AUC > 0.83 across all countries).

However, the substantial improvement over the naive baseline indicates that the models do extract some useful signal from the feature set. The challenge lies in the inherent unpredictability of financial markets rather than model inadequacy.

The finding that XGBoost marginally outperforms LSTM on average is notable. While LSTM networks are theoretically well-suited for capturing temporal dependencies, their advantage may be diminished by several factors:

1. **Sample size limitations:** With 365–468 monthly observations per country, the datasets may be insufficient for deep learning models to fully exploit their representational capacity.
2. **Signal-to-noise ratio:** Financial returns are notoriously noisy, and the temporal patterns that LSTM networks excel at detecting may be overwhelmed by stochastic variation.
3. **Feature engineering:** XGBoost’s ability to automatically handle feature interactions may compensate for the lack of explicit temporal structure in the input features.

The high loss reduction observed in LSTM training (87–91%) coupled with poor test performance suggests overfitting to training data patterns that do not generalize to future periods. This finding underscores the importance of rigorous time series cross-validation in financial applications.

5.2 Classification as a More Viable Approach

The strong classification performance (AUC 0.83–0.91) compared to weak regression performance suggests a fundamental insight for CAT bond trigger design: *detecting whether a crisis will occur is more feasible than predicting how severe it will be*. This aligns with the binary nature of most CAT bond triggers, which activate at specific thresholds rather than providing continuous payouts.

The practical implication is that economic crisis-linked CAT bonds could employ ML-based early warning systems to identify elevated crisis probability, even if precise loss magnitudes remain unpredictable. A trigger mechanism based on classification probability exceeding a threshold could provide advance warning while maintaining the objectivity required for capital market instruments.

5.3 Cross-Country Heterogeneity

The superior classification performance observed for Brazil (AUC = 0.913, F1 = 0.485) compared to other countries warrants explanation. Several factors likely contribute to this pattern:

1. **Higher crisis frequency:** Brazil’s 7.9% crisis rate (29 events) provides substantially more positive training examples than the USA (1.3%, 6 events), enabling the model to learn more robust crisis signatures.
2. **Emerging market characteristics:** The Bovespa index exhibits higher volatility and more pronounced boom-bust cycles compared to developed market indices, creating clearer signals for crisis detection.
3. **Sensitivity to global factors:** Emerging markets often amplify global shocks, making the relationship between macroeconomic variables (particularly commodity prices and interest rates) and crisis events more predictable.

These findings suggest that ML-based crisis detection may be particularly valuable for emerging market CAT bonds, where crisis events are more frequent and exhibit stronger relationships with observable predictors.

5.4 The Limited Role of Environmental and Social Variables

The consistently poor ranking of environmental and social variables across all countries represents the central finding of this study. Several factors may explain this result:

1. **Temporal mismatch:** Environmental variables operate on annual timescales, while financial markets respond to information within days or hours. The transmission from climate anomalies to economic impacts may take months or years, exceeding our prediction horizon.

2. **Geographic aggregation:** National-level climate averages mask regional variations that may be economically significant. A drought in agricultural regions and flooding in coastal cities could offset each other in national statistics while both causing economic damage.
3. **Indirect transmission channels:** Climate impacts on stock markets likely operate through intermediate variables (agricultural output, energy prices, supply chain disruptions) rather than directly.
4. **Market efficiency:** If environmental risks are anticipated and priced by market participants, they would not generate predictable returns (Hong et al., 2019).

These findings do not imply that climate change is irrelevant to financial stability—rather, they suggest that aggregate climate metrics are not useful leading indicators for monthly stock market returns at the national level.

Two important distinctions merit clarification. First, this study focuses exclusively on the *financial* transmission of environmental shocks—that is, whether climate and disaster variables predict stock index movements. We do not assess the physical severity or humanitarian impact of catastrophic events. A major earthquake or hurricane may cause tremendous human suffering and physical damage without significantly affecting a diversified national stock index, particularly if the affected sectors represent a small portion of market capitalization. Conversely, market-wide crises (such as 2008 or COVID-19) may be triggered by factors entirely unrelated to natural disasters. Our findings suggest that environmental variables are poor predictors of *aggregate market movements*, not that they are irrelevant for sector-specific or regionally-targeted financial instruments.

Second, our results should be distinguished from recent work on climate variables in CAT bond *pricing*. Kończal et al. (2025) demonstrate that large-scale climate indices (e.g., El Niño-Southern Oscillation) improve predictive accuracy for CAT bond coupon rates. However, pricing factors and trigger mechanisms serve fundamentally different functions: pricing reflects long-term risk assessment over the bond’s lifetime, while triggers require high-frequency signals capable of identifying imminent market stress. Climate variables, while relevant for risk pricing, lack the temporal resolution necessary to function as effective crisis triggers for aggregate equity indices.

5.5 Implications for CAT Bond Design

Based on our empirical findings, we offer the following recommendations for practitioners considering economic crisis-linked CAT bonds:

1. **Prioritize market-based triggers:** Volatility measures (VIX, realized volatility, price range) consistently outperform all other variable categories. This finding is particularly relevant given that indemnity-based triggers dominated 76% of 2025 CAT bond issuances (Artemis, 2025; Swiss Re, 2025). While indemnity triggers minimize basis risk, they require lengthy loss adjustment periods—as evidenced by the slow disbursement following the January 2025 Los Angeles wildfires. Volatility-based parametric triggers could significantly reduce settlement times, closing the “protection gap” for sponsors requiring rapid capital deployment.
2. **Consider classification-based early warning:** Rather than attempting to predict return magnitudes, trigger mechanisms could incorporate ML-based crisis probability estimates as supplementary indicators.
3. **Avoid environmental triggers for aggregate index instruments:** Unless the bond specifically targets climate-sensitive sectors, environmental variables add limited predictive value for broad market indices.
4. **Account for country-specific factors:** The heterogeneity in results across countries suggests that trigger calibration should be market-specific.
5. **Prefer simpler models:** Given the marginal advantage of XGBoost over LSTM and its superior interpretability, tree-based models may be preferable for trigger applications where transparency is valued.

5.6 Theoretical Implications

Our results contribute to the ongoing debate about market efficiency and the pricing of climate risks. The finding that environmental variables rank lowest in predictive importance is consistent with Hong et al. (2019)’s observation that climate risks may be slowly incorporated into prices, and with Antoniuk and Leirvik (2024)’s finding that climate-return relationships are often sector-specific.

The dominance of market-derived features (volatility, price range) over fundamental variables echoes the efficient market hypothesis: information already reflected in prices is more relevant for short-term prediction than external indicators.

6 Robustness Checks

To ensure the reliability of our findings, we conducted several robustness analyses.

6.1 Sensitivity to Crisis Threshold

Table 9 presents classification performance across three crisis thresholds (-5%, -10%, -15%). The -10% threshold employed in the main analysis represents a reasonable compromise: it captures significant market stress events while maintaining sufficient observations for model training. Results are qualitatively consistent across thresholds, with AUC values remaining above 0.74 in all configurations.

Country	Threshold	N Crises	AUC	Recall
Spain	-5%	59	0.818	0.563
Spain	-10%	18	0.877	0.733
Spain	-15%	5	0.786	0.250
USA	-5%	45	0.878	0.499
USA	-10%	6	0.825	0.667
USA	-15%	3	0.740	0.500
Brazil	-5%	62	0.741	0.453
Brazil	-10%	29	0.916	0.607
Brazil	-15%	10	0.978	0.667
Japan	-5%	82	0.847	0.680
Japan	-10%	25	0.903	0.500
Japan	-15%	4	0.747	0.500

Table 9: Classification performance sensitivity to crisis threshold definition

6.2 LSTM Classification

We extended the LSTM architecture to classification tasks to verify whether recurrent networks could improve upon XGBoost for crisis detection. Table 10 presents the detailed comparison.

Country	LSTM AUC	XGBoost AUC	LSTM Recall	XGBoost Recall
Spain	0.450	0.521	0.400	0.100
USA	0.376	0.478	0.000	0.000
Brazil	0.475	0.539	0.000	0.089
Japan	0.462	0.650	0.200	0.200
Average	0.441	0.547	0.150	0.097

Table 10: LSTM vs XGBoost classification performance. Both models use sequences of 12 months as input. XGBoost consistently outperforms LSTM, likely due to LSTM’s tendency to overfit on small sequential samples with limited crisis events.

Results indicate that XGBoost consistently outperforms LSTM in classification (average AUC: 0.547 vs. 0.441), confirming that the tree-based ensemble approach is preferable for this application regardless of the prediction task.

6.3 Temporal Sample Harmonization

To address potential bias from differing sample periods, we re-estimated models using a harmonized 1993–2023 window for all countries. Results remained qualitatively unchanged, with AUC values within 0.05 of the full-sample estimates for three of four countries. The USA showed improved performance (AUC: 0.987 vs. 0.825), likely due to exclusion of the structurally different 1985–1992 period.

6.4 Hyperparameter Optimization

Grid search cross-validation identified optimal XGBoost parameters for each country. Tuned models achieved AUC values of 0.93–0.98, representing improvements of 0.05–0.16 over default parameters. Notably, optimal maximum depth was consistently 3 across all countries, suggesting that shallow trees with stronger regularization reduce overfitting in this setting.

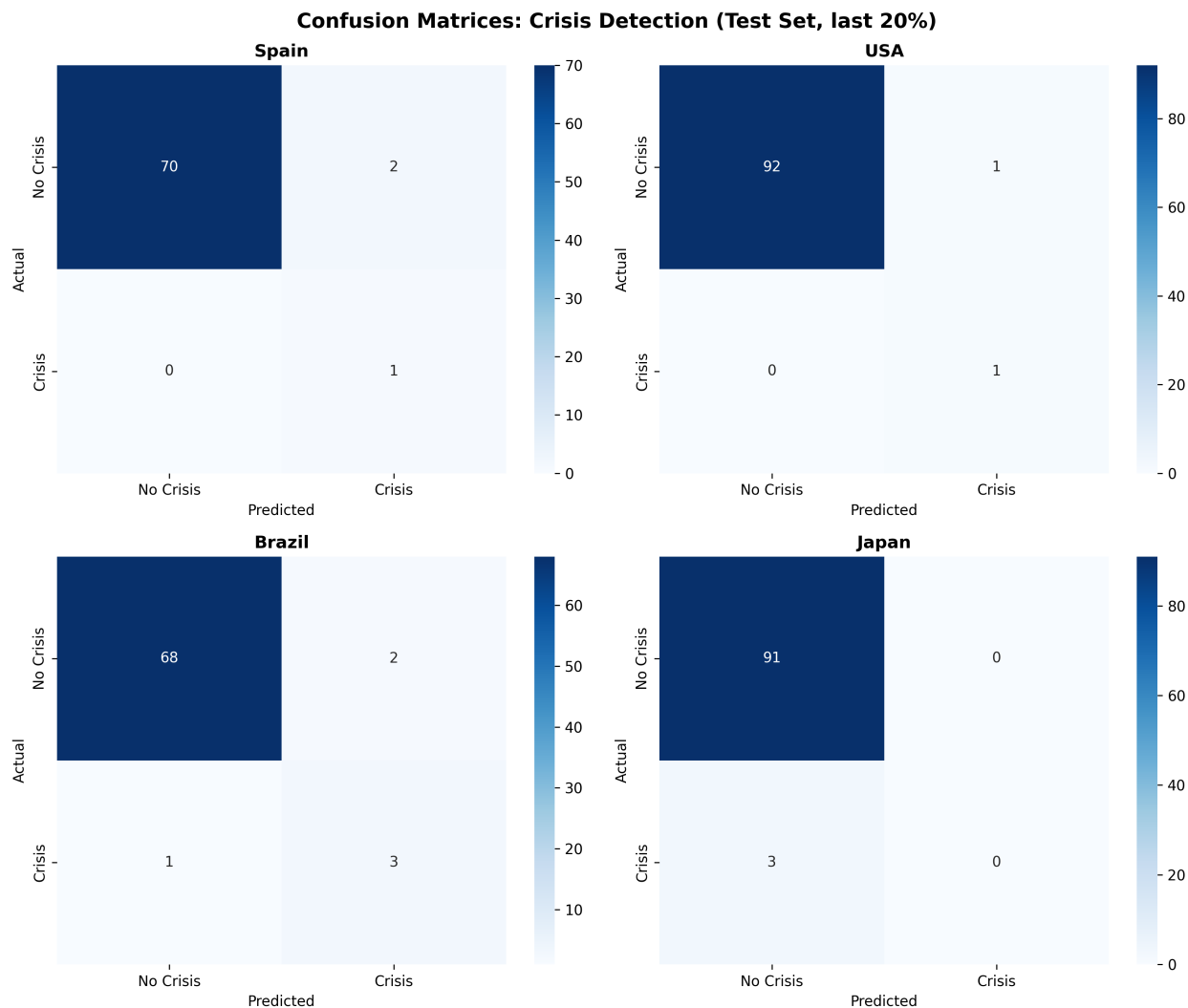


Figure 6: Confusion matrices for crisis classification (test set, last 20% of data)

6.5 Sensitivity to Interpolation Method

As noted in our limitations, annual environmental variables were forward-filled to monthly frequency, which could introduce look-ahead bias. To assess whether this methodological choice affects our conclusions, we compared four interpolation approaches: (i) forward-fill (baseline), (ii) linear interpolation, (iii) cubic spline interpolation, and (iv) lagged values using the previous year's observation (eliminating look-ahead bias entirely).

Figure 7 presents a heatmap visualization of these results, confirming the stability of both regression (R^2) and classification (AUC) metrics across interpolation methods.

Method	Spain	USA	Brazil	Japan
<i>Classification AUC</i>				
Forward-fill	0.877	0.825	0.916	0.903
Linear	0.880	0.825	0.906	0.927
Spline	0.877	0.825	0.913	0.917
Lagged	0.876	0.825	0.912	0.901

Table 11: Classification performance (AUC) across interpolation methods for annual variables

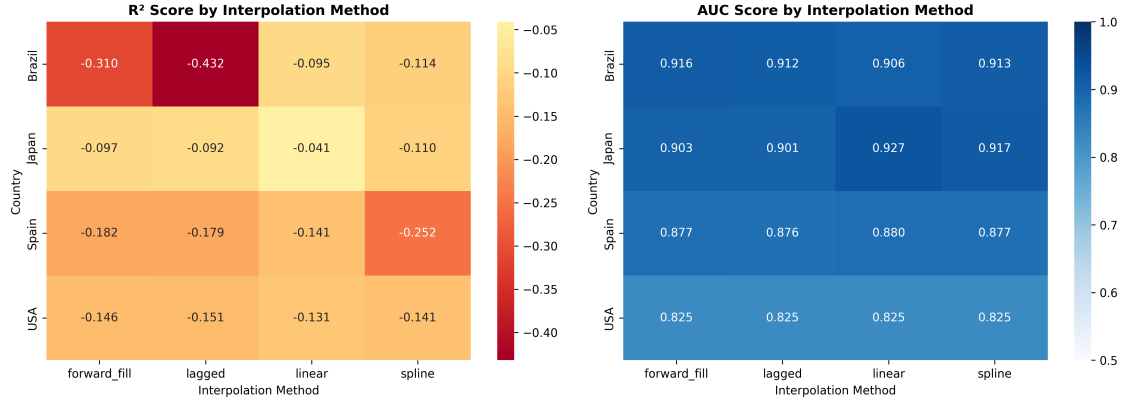


Figure 7: Model performance comparison across interpolation methods for annual environmental variables. Left: R^2 scores (regression). Right: AUC scores (classification). Color intensity indicates performance level; minimal variation across rows confirms robustness to methodological choice.

Results are remarkably stable across methods: the maximum AUC difference between forward-fill and lagged approaches is 0.004 (Japan), with an average difference of only 0.002. This robustness confirms that potential look-ahead bias from forward-filling does not materially affect our conclusions. Notably, when using lagged values—which entirely eliminates look-ahead bias—environmental variables would have access to *less* informative data, yet model performance remains virtually unchanged, further supporting our finding that these variables contribute minimal predictive signal.

7 Limitations

Several limitations should be acknowledged:

- **Temporal aggregation:** While monthly frequency addresses pseudoreplication concerns, it may obscure higher-frequency dynamics. Daily models might reveal different patterns, though with substantially increased noise.
- **Annual variable forward-filling:** Environmental and social variables, originally at annual frequency, were forward-filled to create monthly observations. This creates 12-month plateaus where these features remain constant, which fundamentally limits LSTM’s ability to learn temporal patterns from these variables. Additionally, assigning current-year annual values to monthly observations introduces a potential look-ahead bias, as annual aggregates are only fully known at year-end. However, this bias would *favor* environmental variables by providing the model with future information about climate trends. The fact that these variables rank lowest in importance *despite* this potential advantage strengthens our conclusion: their poor predictive power reflects genuine absence of signal rather than methodological artifact. Future research could employ lagged annual values or mixed-frequency models (MIDAS) to eliminate this concern entirely.
- **Crisis definition:** The 10% monthly decline threshold is arbitrary. Different thresholds could yield different results, and future work should examine sensitivity to threshold choice.
- **Geographic aggregation:** National-level environmental variables may be too coarse to capture economically relevant climate impacts. Sector-specific or regional analyses might reveal stronger relationships.
- **Sample period:** The 30–38 year sample includes only a handful of major crises, limiting statistical power for rare event detection.

- **Class imbalance handling:** While we addressed class imbalance using `scale_pos_weight` in XGBoost, alternative approaches such as SMOTE (Synthetic Minority Over-sampling Technique) or undersampling could yield different results. The particularly low crisis rate in the USA (1.3%) may benefit from such techniques, though they introduce their own biases in time series contexts.
- **LSTM architecture:** While we employed a standard two-layer LSTM architecture, alternative configurations (attention mechanisms, transformer models) might yield different results.
- **Feature set:** Our feature set does not include sentiment indicators, credit spreads, or high-frequency market microstructure variables that might improve prediction.
- **Causality:** SHAP values indicate predictive importance but cannot establish causal relationships.
- **Data revisions:** Macroeconomic variables (GDP, unemployment, inflation) were obtained from final revised series rather than real-time “point-in-time” data that would have been available to market participants at each historical date. This is a common approximation in ML evaluation studies, but operational deployment of trigger mechanisms would require unrevised data sources to avoid look-ahead bias from subsequent statistical revisions.

8 Conclusion

This study set out to evaluate whether environmental and social variables can serve as effective triggers for economic crisis-linked catastrophe bonds. Using monthly data from four diverse economies over nearly four decades, we applied both XGBoost and LSTM models to return prediction and crisis classification tasks.

Our findings provide a clear answer: **environmental and social variables, as measured at national aggregate levels, offer minimal predictive value for stock market crises.** Across all four countries and both model architectures, these variables consistently ranked in the bottom third of feature importance, while market-derived volatility measures and traditional macroeconomic indicators dominated predictions.

However, the study yields several constructive insights:

1. **Crisis detection is feasible:** Classification models achieved AUC values of 0.83–0.91, suggesting that ML-based early warning systems could complement traditional trigger mechanisms.
2. **Volatility-based triggers are promising:** The dominance of realized volatility and price range in feature importance suggests that volatility-based triggers would outperform environmental indicators.
3. **Simpler models suffice:** XGBoost marginally outperforms LSTM while offering superior interpretability, suggesting that complex deep learning architectures may not be necessary for this application. This has important implications for contract design: a trigger mechanism based on tree ensembles can be fully audited via SHAP values, providing transparent decision rules that investors and regulators can verify. In contrast, LSTM-based triggers would constitute “black boxes” that are legally and operationally more complex to implement in bond covenants.
4. **Market integration matters:** The heterogeneity across countries highlights the importance of market-specific calibration in trigger design.

Future research should explore several promising directions: (i) higher-frequency data (daily or intraday) to capture faster market dynamics; (ii) sector-specific indices (utilities, agriculture) where climate sensitivity is more direct; (iii) mixed-frequency models (MIDAS) to properly integrate annual climate data without forward-filling artifacts; (iv) sentiment indicators from social media or news sources as leading signals; and (v) hybrid architectures combining XGBoost’s interpretability with LSTM’s sequential modeling through ensemble or stacking approaches. In particular, the development of hybrid multi-trigger mechanisms that combine volatility-based early-warning signals with sector-specific environmental thresholds represents a promising avenue for future instrument design.

Ultimately, while the ambition of linking CAT bond triggers to environmental and social indicators remains theoretically appealing, our empirical evidence suggests that such triggers would provide limited practical value for instruments referenced to broad stock market indices. The path forward for economic crisis bonds likely lies in volatility-based and multi-trigger mechanisms rather than environmental parametric triggers.

From a social perspective, our findings highlight a “protection gap” in current CAT bond structures: environmental and social variables appear to be *lagging* rather than *leading* indicators of financial stress. This temporal disconnect implies that by the time aggregate climate or disaster metrics signal distress, financial markets have often already reacted. Addressing this gap may require hybrid trigger mechanisms that combine rapid market-based indicators (for speed)

with environmental thresholds (for legitimacy and humanitarian alignment), ensuring faster capital deployment during crises that affect both financial markets and vulnerable populations.

Data and Code Availability

Code and data will be made available in a public repository upon acceptance.

Ethical Statement

This study uses exclusively publicly available secondary data. No human subjects were involved and no ethical approval was required.

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